Sixth Section.

Multiplication and Division of the Elliptic Functions.

§ 63. The Galois group of the division equation.

In the thirteenth section of the first volume it was shown how the algebraic nature of an equation finds its simplest expression in the Galois group of the equation.

Let us briefly recall here the definition of the Galois group.

After the domain of rationality Ω has been fixed, the Galois group of an equation $F(x) = 0$, of which it is only assumed that it has no equal roots, is defined as the group $\mathfrak{G}$ of the permutations among the roots of this equation which has the following double property:

a) Every rational equation in Ω which holds between the roots of $F(x)$ remains true when the roots are subjected to any permutation of this group.

b) Every rational function in Ω of the roots of $F(x)$ which admits all permutations of the group is contained in Ω.

Since we now seek to determine the group $\mathfrak{G}$ of the division equation, we first assume as domain of rationality the totality of all rational numbers and rational functions of $\varkappa^2$.

By the addition theorem and the multiplication theorem (compare § 57, 61), if f and φ_m denote rational functions which are independent of μ, μ', ν, ν', one has

$$x_{\mu+\nu,\mu'+\nu'} = f(x_{\mu,\mu'}, x_{\nu,\nu'}), \tag{1}$$

$$x_{m\mu,m\mu'} = \varphi_m(x_{\mu,\mu'}). \tag{2}$$

Let now S be any substitution of the group $\mathfrak{G}$, by which, if a, b, c, ∂ are integers taken modulo n,

$$x_{1,0},\ x_{0,1} \quad \text{go into} \quad x_{\partial,-c},\ x_{-b,a}.$$

Then we may apply S to each of the formulae (1), (2). But by (2)

$$x_{\mu,0} = \varphi_\mu(x_{1,0}),$$

and from this it follows that under the substitution S

$$x_{\mu,0} \quad \text{goes into} \quad \varphi_\mu(x_{\partial,-c}) = x_{\partial\mu,-c\mu}.$$

Similarly one finds that under S

$$x_{0,\mu'} \quad \text{goes into} \quad \varphi_{\mu'}(x_{-b,a}) = x_{-b\mu',a\mu'}.$$

Applying this to the equation following from (1),

$$x_{\mu,\mu'} = f(x_{\mu,0}, x_{0,\mu'}),$$

one obtains that under S

$$x_{\mu,\mu'} \quad \text{goes into} \quad x_{\partial\mu-b\mu',-c\mu+a\mu'}. \tag{3}$$

From this we first conclude that the numbers a, b, c, ∂ must be so constituted that their determinant

$$m = a\partial - bc \tag{4}$$

has no divisor in common with n. For if such a divisor were present, then μ, μ' could be so chosen, without both being divisible by n, that

$$\partial\mu - b\mu' \equiv 0, \qquad -c\mu + a\mu' \equiv 0 \pmod n,$$

and several distinct roots $x_{\mu,\mu'}$ would pass under S into one and the same root, which is impossible. For first, neither a and b, nor c and ∂, can have a divisor in common with n, since otherwise, by taking $a\mu', b\mu'$, or $c\mu, \partial\mu$, divisible by n, all $x_{\mu',0}$ or all $x_{0,\mu}$ would pass into $x_{0,0}$. If one then puts

$$m\mu = n(ha + h'b), \qquad m\mu' = n(hc + h'\partial),$$

and determines h and h' so that $ha + h'b$ has no divisor in common with n, then, when n and m have a common divisor, μ, μ' need not both be divisible by n, and not only $x_{0,0}$ but also $x_{\mu,\mu'}$ passes, by (3), into $x_{0,0}$.

If we denote the interchange (3), in abbreviated form, by

$$(x_{\mu,\mu'}, x_{\nu,\nu'}),$$

then

$$\nu \equiv \partial\mu - b\mu', \qquad \nu' \equiv -c\mu + a\mu' \pmod{n},$$

and from this, by solving, using the notation of § 28, (4),

$$m(\mu, \mu') \equiv \begin{pmatrix} a & b \\ c & \partial \end{pmatrix} (\nu, \nu'). \tag{5}$$

It is therefore

$$\begin{pmatrix} a & b \\ c & \partial \end{pmatrix} \tag{6}$$

a convenient sign for the interchange S, and from (5) one sees that two such interchanges

$$S = \begin{pmatrix} a & b \\ c & \partial \end{pmatrix}, \qquad S' = \begin{pmatrix} a' & b' \\ c' & \partial' \end{pmatrix}$$

are composed exactly according to the rule given in § 28:

$$SS' = S'' = \begin{pmatrix} aa' + bc' & ab' + b\partial' \\ ca' + \partial c' & cb' + \partial\partial' \end{pmatrix}, \tag{7}$$

so that the interchange S'' of the roots arises when first the interchange S, and then the interchange S', is performed among the roots of the division equation. It must always be observed, however, that here only the residues modulo n of the numbers a, b, c, ∂ come into account, so that the number of the interchanges S is always finite.[1]

1. The totality of all substitutions $\begin{pmatrix} a & b \\ c & \partial \end{pmatrix}$ forms a group, which we denote by $\mathfrak{A}$, and in it, as we have proved, the group $\mathfrak{G}$ of the division equation is contained.

In $\mathfrak{A}$ there is contained, as a divisor, a group $\mathfrak{B}$, consisting of all substitutions

$$T = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \tag{8}$$

which satisfy the condition

$$\alpha\delta - \beta\gamma \equiv 1 \pmod{n}. \tag{9}$$

Every substitution S can be derived by composition of

$$\begin{pmatrix} m & 0 \\ 0 & 1 \end{pmatrix}$$

[1] If, as at first sight might seem more natural, one chose the notation $S = \begin{pmatrix} \partial & -b \\ -c & a \end{pmatrix}$, the composition would have to be denoted by the formula $S'S = S''$, hence conversely to the usual composition of permutations.

with a substitution T; for, in order that

$$S = \begin{pmatrix} m & 0 \\ 0 & 1 \end{pmatrix} T,$$

$\alpha, \beta, \gamma, \delta$ are simply to be determined from the congruences

$$a \equiv \alpha m, quad b \equiv \beta m, \qquad c \equiv \gamma, \qquad \partial \equiv \delta \pmod{n}.$$

We now consider

$$x_{\mu,\mu'} = \operatorname{sn}\left(\frac{4\mu K + 4\mu' \mathrm{i} K'}{n}\right) = \frac{\vartheta_{00}\,\vartheta_{11}\left(\frac{2\mu+2\mu'\omega}{n}\right)}{\vartheta_{10}\,\vartheta_{01}\left(\frac{2\mu+2\mu'\omega}{n}\right)}, \qquad \varkappa^2 = \frac{\vartheta_{10}^4}{\vartheta_{00}^4}, \tag{10}$$

as functions of ω, and apply to them a linear transformation

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \equiv \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \pmod{8}$$

by replacing ω by

$$\omega' = \frac{\gamma + \delta\omega}{\alpha + \beta\omega}. \tag{11}$$

Then from the formulae § 39, (3), (5), and (6), it follows that

$$x_{\mu,\mu'} \quad \text{goes into} \quad x_{\alpha\mu+\gamma\mu',\beta\mu+\delta\mu'},$$

that is, the $x_{\mu,\mu'}$ undergo a substitution which, by (3), (6), is to be denoted by

$$T = \begin{pmatrix} \delta & -\gamma \\ -\beta & \alpha \end{pmatrix},$$

while $\varkappa^2$ remains unchanged.

In this way, conversely, every one of the substitutions T can arise; for if any substitution T is given, then by adding suitable multiples of the odd number n to $\alpha, \beta, \gamma, \delta$ one can always satisfy the conditions

$$\alpha \equiv 1, \qquad \delta \equiv 1, \qquad \beta \equiv 0, \qquad \gamma \equiv 0 \pmod{8}.$$

Any rational equation between the roots $x_{\mu,\mu'}$ and $\varkappa^2$, even if arbitrary constant, that is, ω-independent, numerical coefficients occur in it, passes, by (10), into an identity; and one may therefore substitute for ω the expression

$$\frac{\gamma + \delta\omega}{\alpha + \beta\omega},$$

that is, one may apply every substitution T to the rational equation between the $x_{\mu,\mu'}$. Hence it follows that the whole group $\mathfrak{B}$ is contained in $\mathfrak{G}$ (Vol. I, § 156), and even in the group $\mathfrak{G}'$ which arises from the group $\mathfrak{G}$ of the division equation by adjunction of arbitrary numbers.

If we enlarge the domain of rationality of the rational numbers by adjoining the nth roots of unity, then the Abelian relations of the preceding paragraph also belong to the rational equations. But to these relations, for instance to

$$\sum^{\nu} e^{\frac{8\nu'\nu\pi\mathrm{i}}{n}} x_{\nu,\nu'} = 0,$$

none of the substitutions

$$M = \begin{pmatrix} m & 0 \\ 0 & 1 \end{pmatrix}$$

in which m differs from 1 is applicable. For by this substitution, if $mm' \equiv 1 \pmod{n}$, there passes

$$\sum^{\nu} e^{\frac{8\nu'\nu\pi i}{n}} x_{\nu,\nu'} \quad \text{into} \quad \sum^{\nu} e^{\frac{8m'\nu'\nu\pi i}{n}} x_{\nu,\nu'},$$

which by § 62, (12), is different from zero if m' and therefore also m is not congruent to 1 modulo n. Thus the following theorem results:

2. After adjunction of the nth roots of unity (and of arbitrary other constants), $\mathfrak{B}$ is the Galois group of the division equation. It will also be called the monodromy group of the division equation.

3. We shall now still prove that, in the original domain of rationality, hence without adjunction of the nth roots of unity or of other irrational numbers, the group of the division equation is identical with the group $\mathfrak{A}$.

It has been shown in Vol. I, § 174, that the primitive nth roots of unity ρ are roots of an irreducible equation

$$\Phi(\rho) = 0, \tag{12}$$

and that this equation also cannot decompose when the independent variable $\varkappa^2$ is adjoined to the domain of rationality of the rational numbers.

The degree of this equation is $\varphi(n)$, that is, equal to the number of residues modulo n which are relatively prime to n.

Let now

$$F(r) = 0 \tag{13}$$

be a Galois resolvent of degree μ of the division equation in the original domain of rationality (so that all roots $x_{\nu,\nu'}$ are rationally representable by r, Vol. I, § 152). This equation must decompose after adjunction of a root of (12); for if this were not so, every rational relation

$$\psi(r, \rho) = 0 \tag{14}$$

would be satisfied for all roots of (13), and $\psi(t, \rho)$ would be divisible by $F(t)$, for a variable t. The same would still hold if ρ were replaced by any other root ρ^m of (12). This, however, is not the case by § 62, (12), if in place of (14) one puts one of the Abelian relations.

Let now, after adjunction of ρ,

$$F(r, \rho) = 0 \tag{15}$$

be the Galois resolvent of degree ν of the division equation, so that ν is, by 2., equal to the degree of the group $\mathfrak{B}$. Then $F(t)$ is algebraically divisible by $F(t, \rho)$, and therefore, by the irreducibility of (12), also by each of the functions $F(t, \rho^m)$. Since the functions $F(t, \rho^m)$ are irreducible, two of them can have a common divisor only when they are entirely identical. But if

$$F(t, \rho) = F(t, \rho^m)$$

were true, then it would follow from every equation of the form (14) that $\psi(t, \rho)$ is divisible by $F(t, \rho) = F(t, \rho^m)$, and it would follow that in (14) the interchange (ρ, ρ^m) is admissible, which, again by the Abelian relations, is not the case. Consequently the $\varphi(n)$ functions $F(t, \rho^m)$ are all different from one another, and $F(t)$ is divisible by their product. Thus the degree μ of $F(t)$ is at least equal to $\nu\varphi(n)$. But it also cannot be larger than $\nu\varphi(n)$, since $\nu\varphi(n)$ is the degree of the group $\mathfrak{A}$, and since the group $\mathfrak{G}$, of degree μ, is surely contained in $\mathfrak{A}$. It follows from this that

$$\mu = \nu\varphi(n), \tag{16}$$

and at the same time that $\mathfrak{G}$ is identical with $\mathfrak{A}$.

But from this it also follows that $F(t)$ is equal to the product of all factors $F(t, \rho^m)$, hence

$$F(t) = \prod^{m} F(t, \rho^m), \tag{17}$$

and, since $F(t) = 0$ has no equal roots, that $F(r, \rho)$ indeed vanishes, but none of the other factors $F(r, \rho^m)$. The equations

$$F(r, t) = 0, \qquad \Phi(t) = 0 \tag{18}$$

therefore have only the one root $t = \rho$ in common, and by seeking their greatest common divisor one finds ρ rationally expressed by r, that is, by the roots of the division equation.

These expressions for ρ do not change their value when a substitution of the group $\mathfrak{B}$ is applied to r, whereas, by (12), ρ passes by a substitution of $\mathfrak{A}$ into a power of ρ.

The Abelian relations show that under the substitution contained in $\mathfrak{A}$

$$\begin{pmatrix} m & 0 \\ 0 & 1 \end{pmatrix} \tag{19}$$

ρ goes into ρ^m. For if we put

$$e^{\frac{8\pi i}{n}} = \rho^\lambda,$$

one of the Abelian relations is

$$\sum^{\nu} \rho^{\lambda\nu\nu'} x_{\nu,\nu'} = 0, \tag{20}$$

and, when the expression through r is put for ρ, all substitutions of $\mathfrak{A}$, hence also (19), are applicable. By § 62, (12), however, (20) remains correct under this substitution only when ρ passes into ρ^m.

Since ρ is not changed by the substitutions in $\mathfrak{B}$, and since the group $\mathfrak{A}$ arises from $\mathfrak{B}$ by composition with (19), it follows that under any substitution in $\mathfrak{A}$

$$\begin{pmatrix} a & b \\ c & \partial \end{pmatrix}$$

ρ passes into ρ^m, where m is congruent to the determinant $(a\partial - bc)$ modulo n.

4. We finally determine still the number ν, that is, the degree of the group $\mathfrak{B}$.

As is known, $\varphi(n)$ has the expression (Vol. I, § 140)

$$\varphi(n) = n \prod \left(1 - \frac{1}{p}\right), \tag{21}$$

where the product sign extends over all distinct prime numbers p contained in n.

The number ν is equal to the number of systems of numbers $\alpha, \beta, \gamma, \delta$, not congruent modulo n, which satisfy the condition

$$\alpha\delta - \beta\gamma \equiv 1 \pmod{n}. \tag{22}$$

We ask first for the number of pairs α, β which have no common divisor with n, and denote it by $\chi(n)$.

If $n = n'n''$ and n' is relatively prime to n'', then from every combination of a number-pair α', β' belonging to n' with a number-pair α'', β'' belonging to n'' one can derive a pair belonging to n by

$$\alpha = n''\alpha' + n'\alpha'', \qquad \beta = n''\beta' + n'\beta'',$$

and in this way one obtains all number-pairs α, β, and each only once. Hence

$$\chi(n) = \chi(n')\chi(n''). \tag{23}$$

It remains to determine $\chi(p^\pi)$, that is, $\chi(n)$ for a prime-power $n = p^\pi$.

If one first sets for α, β all numbers distinct modulo p^π, then their number is $p^{2\pi}$. But one must subtract all those pairs in which both α and β are divisible by p; their number is $p^{2\pi-2}$. Thus

$$\chi(p^\pi) = p^{2\pi} - p^{2\pi-2},$$

and consequently, by (23), in general

$$\chi(n) = n^2 \prod \left(1 - \frac{1}{p^2}\right), \tag{24}$$

where p runs through the prime numbers contained in n.

Each number-pair α, β can be transformed, by increasing by multiples of n, into one whose numbers are relatively prime among themselves, and then γ, δ can be determined so that

$$\alpha\delta - \beta\gamma = 1.$$

Herein γ, δ can be replaced by $\gamma + h\alpha, \delta + h\beta$, and letting h run through a complete residue system modulo n shows that to each number-pair α, β there belong n number-pairs γ, δ satisfying (22). Consequently the order of the group $\mathfrak{B}$ is

$$\nu = n^3 \prod \left(1 - \frac{1}{p^2}\right),$$

or, if we introduce also the numerical function

$$\psi(n) = n \prod \left(1 + \frac{1}{p}\right), \tag{25}$$

then

$$\nu = n\varphi(n)\psi(n), \tag{26}$$

and the order of the group $\mathfrak{A}$ is

$$\mu = n\varphi(n)^2\psi(n). \tag{27}$$

§ 64. The irreducible factors of the division equation.

If

$$\nu = \partial\mu - b\mu', \qquad \nu' = -c\mu + a\mu',$$

and $a\partial - bc$ is relatively prime to n, then the greatest common divisor of μ, μ', n is at the same time the greatest common divisor of ν, ν', n. Thus the roots $x_{\mu,\mu'}$ of the division equation decompose, according to the greatest common divisor of μ, μ', n, into systems which can only ever pass into one another by the substitutions of the group $\mathfrak{A}$; that is, the group $\mathfrak{G}$ is intransitive, and the systems of intransitivity are the systems $x_{\mu,\mu'}$ in which μ, μ', n have one and the same greatest common divisor. The division equation is therefore reducible (except when n is a prime number), and the roots of one of these systems of imprimitivity satisfy a rational equation (Vol. I, § 157).

By the preceding paragraph there are $\varphi(n)\psi(n)$ number-pairs μ, μ' whose greatest common divisor with n is equal to 1, and the roots $x_{\mu,\mu'}$ corresponding to these number-pairs therefore satisfy a rational equation of degree $\varphi(n)\psi(n)$. We call this equation the proper division equation for the divisor n, because only when μ, μ', n have no common divisor is $x_{\mu,\mu'}$ not at the same time a root of a division equation for a smaller divisor. In contrast to this, we call the equation whose roots are all of the $x_{\mu,\mu'}$ the general division equation for the divisor n.

Under any substitution of the group $\mathfrak{A}$

$$S = \begin{pmatrix} a & b \\ c & \partial \end{pmatrix},$$

$(1, 0), (0, 1)$ pass into $(\partial, -c), (-b, a)$; and if S is not the identical substitution, then certainly at least one of the two roots $x_{1,0}, x_{0,1}$ is changed by S. It follows from this that the Galois group of the proper division equation is exactly the same as that of the general one, only applied to the case in which μ, μ', n have no common divisor, namely, according as the nth roots of unity are adjoined or not, $\mathfrak{B}$ or $\mathfrak{A}$.

From this it follows further that the proper division equation is irreducible, even after adjunction of arbitrary constants.

For if γ, δ are any two numbers having no common divisor with n, then one can determine α, β according to the congruence

$$\alpha\delta - \beta\gamma \equiv 1 \pmod{n},$$

and apply the substitution contained in $\mathfrak{B}$

$$T = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$$

to every rational equation between the roots $x_{\mu,\mu'}$. Thus if $x_{1,0}$ satisfies a rational equation, with arbitrary constant coefficients,

$$\Phi(x_{1,0}) = 0,$$

then every other root $x_{\delta,-\gamma}$ of the proper division equation satisfies the same equation, and from this the irreducibility of the latter follows.

§ 65. Reduction of the division equation to transformation equations.

The roots of the proper division equation can be arranged in rows in the following manner. Choose arbitrarily one of the roots

$$x_{\mu_1,\mu_1'} = \operatorname{sn}\left(\frac{4\mu_1 K + 4\mu_1' \mathrm{i} K'}{n}\right) = \operatorname{sn}\Omega_1.$$

Among the roots of the division equation there also occur the $\varphi(n)$ quantities

$$(R_1) \qquad \operatorname{sn} h\Omega_1,$$

which, when h runs through a complete system of incongruent numbers prime to n, are all different from one another. We shall call the system (R_1) the first row of the roots.

If now $\operatorname{sn}\Omega_2$ is a root not contained in (R_1), then the $\varphi(n)$ quantities

$$(R_2) \qquad \operatorname{sn} h\Omega_2,$$

which are different both among themselves and from the roots (R_1), form a second row; and in this way one can continue until all $\varphi(n)\psi(n)$ roots of the proper division equation have been distributed into $\psi(n)$ rows of $\varphi(n)$ terms each.

By the multiplication theorem every other root of the same row can be rationally expressed through one root in the form

$$\operatorname{sn} h\Omega = f_h(\operatorname{sn}\Omega), \tag{1}$$

where f_h is a rational function depending only on the multiplier h, and not on the choice of Ω.

If the two roots $x_{\mu,\mu'}$, $x_{\nu,\nu'}$ belong to the same row, then h must be choosable so that

$$h\mu \equiv \nu, \qquad h\mu' \equiv \nu' \pmod{n}, \tag{2}$$

and conversely, if this is the case, the two roots belong to the same row. But from (2) the congruence

$$\mu\nu' - \nu\mu' \equiv 0 \pmod{n} \tag{3}$$

follows, and conversely the congruences (2) can again be derived from this.

For since μ, μ', n are relatively prime, one can determine α, β so that

$$\alpha\mu - \beta\mu' \equiv 1 \pmod{n},$$

and from this, with the help of (3), it follows that

$$\nu \equiv (\alpha\nu - \beta\nu')\mu, \qquad \nu' \equiv (\alpha\nu - \beta\nu')\mu' \pmod{n}.$$

Thus, if $\alpha\nu - \beta\nu' = h$ is put, the congruences (2) result.

The congruence (3) is therefore the necessary and sufficient condition that the two roots $x_{\mu,\mu'}$, $x_{\nu,\nu'}$ of the proper division equation belong to the same row.

It follows from this that the division into rows is entirely independent of the arbitrariness in the assumption about the roots $\operatorname{sn}\Omega_1, \operatorname{sn}\Omega_2, 7\ldots$.

But still more follows from this, namely that the substitutions of the group $\mathfrak{A}$ do not tear the rows apart, but only interchange them among one another. The group of the division equation is therefore imprimitive, and the individual rows are the systems of imprimitivity (Vol. I, § 158).

For if one applies simultaneously to (μ, μ') and (ν, ν') the substitution

$$\begin{pmatrix} a & b \\ c & \partial \end{pmatrix},$$

then the congruence (3) remains preserved.

If one selects, among the substitutions of the group $\mathfrak{A}$, those substitutions which only interchange among themselves the roots of a row, one obtains a group, and indeed a divisor of $\mathfrak{A}$. Thus to every row there belongs such a divisor of $\mathfrak{A}$. Let $R_{\mu,\mu'}$ denote the row in which the root $x_{\mu,\mu'}$ occurs, and let $\mathfrak{A}_{\mu,\mu'}$ denote the divisor of $\mathfrak{A}$ belonging to this row. Then, by (3) [compare § 63, (3)], the substitutions

$$\begin{pmatrix} a & b \\ c & \partial \end{pmatrix}$$

of $\mathfrak{A}_{\mu,\mu'}$ are characterized by the congruence

$$(\partial\mu - b\mu')\mu' + (c\mu - a\mu')\mu \equiv 0 \pmod{n}. \tag{4}$$

Similarly one obtains a group $\mathfrak{B}_{\mu,\mu'}$ if one seeks the substitutions in $\mathfrak{B}$ which only interchange among themselves the roots of the row $R_{\mu,\mu'}$; their substitutions

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$$

are characterized by the two congruences

$$\begin{matrix} (\delta\mu - \beta\mu')\mu' + (\gamma\mu - \alpha\mu')\mu \equiv 0, \\ \alpha\delta - \beta\gamma \equiv 1 \end{matrix} \quad \pmod{n}. \tag{5}$$

For example, the groups $\mathfrak{A}_{1,0}$, $\mathfrak{B}_{1,0}$ consist of the substitutions

$$\begin{pmatrix} a & b \\ 0 & \partial \end{pmatrix}, \qquad \begin{pmatrix} \alpha & \beta \\ 0 & \delta \end{pmatrix}, \qquad \alpha\delta \equiv 1 \pmod{n}.$$

The groups $\mathfrak{A}_{\mu,\mu'}$ are conjugate divisors of $\mathfrak{A}$ (Vol. II, § 3), for if $x_{1,0}$ passes under S into $x_{\mu,\mu'}$, and hence $R_{1,0}$ into $R_{\mu,\mu'}$, then under the group

$$S^{-1}\mathfrak{A}_{1,0}S$$

the roots of $R_{\mu,\mu'}$ are interchanged only among themselves, and hence

$$S^{-1}\mathfrak{A}_{1,0}S = \mathfrak{A}_{\mu,\mu'}. \tag{6}$$

Likewise, if T is a substitution in $\mathfrak{B}$ under which $x_{1,0}$ passes into $x_{\mu,\mu'}$, then

$$T^{-1}\mathfrak{B}_{1,0}T = \mathfrak{B}_{\mu,\mu'}. \tag{7}$$

The greatest common divisor $\mathfrak{A}_0$ of all the groups $\mathfrak{A}_{\mu,\mu'}$ is determined by (4) as

$$b \equiv 0, \qquad c \equiv 0, \qquad a \equiv \partial \pmod n,$$

and therefore consists of the substitutions

$$\begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix},$$

where a is any number prime to n. $\mathfrak{A}_0$ is identical with the greatest common divisor of three of the groups $\mathfrak{A}_{\mu,\mu'}$ which belong to different rows. For $\mathfrak{A}_0$ is the greatest common divisor of $\mathfrak{A}_{\mu,\mu'}, \mathfrak{A}_{\nu,\nu'}, \mathfrak{A}_{\rho,\rho'}$ when the three congruences

$$\begin{aligned} c\mu^2 - (\partial - a)\mu\mu' - b\mu'^2 &\equiv 0, \\ c\nu^2 - (\partial - a)\nu\nu' - b\nu'^2 &\equiv 0, \quad \pmod n \\ c\rho^2 - (\partial - a)\rho\rho' - b\rho'^2 &\equiv 0 \end{aligned}$$

are fulfilled only under the assumption

$$\partial \equiv a, \qquad b \equiv 0, \qquad c \equiv 0 \pmod n.$$

This occurs when the determinant

$$\begin{vmatrix} \mu^2 & \mu\mu' & \mu'^2 \\ \nu^2 & \nu\nu' & \nu'^2 \\ \rho^2 & \rho\rho' & \rho'^2 \end{vmatrix} = -(\nu\rho' - \rho\nu')(\rho\mu' - \mu\rho')(\mu\nu' - \nu\mu')$$

is relatively prime to n.

Similarly, by (5), the greatest common divisor $\mathfrak{B}_0$ of all $\mathfrak{B}_{\mu,\mu'}$ is the totality of the substitutions

$$\begin{pmatrix} \alpha & 0 \\ 0 & \alpha \end{pmatrix} \tag{8}$$

with the condition

$$\alpha^2 \equiv 1 \pmod n. \tag{9}$$

The congruence (9) has, if k is the number of distinct prime numbers contained in n, 2^k incongruent solutions, and this is therefore the degree of the group $\mathfrak{B}_0$.

A rational function ξ of the roots of one row, say the row $R_{1,0}$, can by (1) be represented rationally by one of these roots. If this function has the property of remaining unchanged when this one root is replaced by another of the same row, for example if ξ is a symmetric function of the roots of one row, then ξ obtains, under application of the substitutions of $\mathfrak{A}$ (or also of $\mathfrak{B}$), only

$$\nu = \psi(n) \tag{10}$$

different values

$$\xi_1, \xi_2, \ldots, \xi_\nu; \tag{11}$$

and if these values are all different from one another, then ξ belongs to the group $\mathfrak{A}_{1,0}$, and all other symmetric functions of the roots of $R_{1,0}$ are rationally representable by ξ (Vol. I, § 162).

The ν quantities (11) are the roots of an irreducible rational equation of the νth degree, which we call a transformation equation.

If only rational numerical coefficients occur in ξ, the transformation equation itself has rational numerical coefficients. It remains irreducible, however, even when nth roots of unity, or indeed any constants whatever, are adjoined; this follows from the fact that by substitutions of the group $\mathfrak{B}$ every root of the division equation can be carried into every other, and hence every row into every other row.

By adjunction of one root ξ of the transformation equation, say the one belonging to the group $\mathfrak{A}_{1,0}$, the group of the division equation is reduced to $\mathfrak{A}_{1,0}$. This latter group, however, is no longer transitive, but interchanges among themselves the $\varphi(n)$ roots $x_{h,0}$, where h runs through a complete system of incongruent numbers prime to n. The division equation therefore becomes reducible and has a factor of degree $\varphi(n)$, whose roots are the $x_{h,0}$. The influence of a substitution of the group $\mathfrak{A}_{1,0}$,

$$\begin{pmatrix} a & b \\ 0 & \partial \end{pmatrix},$$

on $x_{h,0}$ consists in $x_{h,0}$ passing into $x_{ah,0}$; and the group of this equation of degree $\varphi(n)$ therefore consists of the interchanges

$$(x_{h,0}, x_{ah,0}),$$

where a may be any number prime to n. This group is Abelian, and therefore the roots $x_{h,0}$ are algebraically determinable by radicals after adjunction of ξ.

If, however, we adjoin not merely one, but all roots ξ of the transformation equation, or, what amounts to the same thing, the three belonging to $R_{1,0}, R_{0,1}, R_{1,1}$, then the group of the division equation is reduced to $\mathfrak{A}_0$; and if we still adjoin nth roots of unity, to $\mathfrak{B}_0$. The group $\mathfrak{B}_0$ is an Abelian group of degree 2^k, containing only elements whose degree is 2. Consequently the division equation is solvable by square roots.

In order to find the form of this solution, put

$$n = p^{\pi} p'^{\pi'} \cdots,$$

where $p, p', \dots$ are distinct prime numbers and $\pi, \pi', \dots$ positive exponents. Determine the numbers $c, c', \dots$ from the congruences

$$\begin{aligned} c &\equiv -1 \pmod{p^{\pi}}, & c' &\equiv -1 \pmod{p'^{\pi'}}, \dots, \\ c &\equiv +1 \pmod{np^{-\pi}}, & c' &\equiv +1 \pmod{np'^{-\pi'}}, \dots. \end{aligned}$$

Then every solution α of the congruence

$$\alpha^2 \equiv 1 \pmod{n}$$

is obtained, and each only once, in the form

$$\alpha \equiv c^{\varepsilon} c'^{\varepsilon'} \cdots \pmod{n}, \tag{12}$$

where $\varepsilon, \varepsilon', \dots$ take the values $0, 1$.

If now Ω is any one of the values

$$\frac{4\mu K + 4\mu' \mathrm{i} K'}{n},$$

so that $\operatorname{sn}\Omega$ is any root of the proper division equation, then the sum, extended over all α,

$$\sum (\pm 1)^{\varepsilon} (\pm 1)^{\varepsilon'} \cdots \operatorname{sn} \alpha\Omega = \psi(\Omega), \tag{13}$$

in which the k signs ± 1 can be chosen arbitrarily, has the property that for every α determined by (12),

$$\psi(\alpha\Omega) = (\pm 1)^{\varepsilon}(\pm 1)^{\varepsilon'} \cdots \psi(\Omega), \tag{14}$$

and the square of $\psi(\Omega)$ remains unchanged by the substitutions of the group $\mathfrak{B}_0$, hence is rationally expressible by nth roots of unity and the roots of a transformation equation. Thus $\psi(\Omega)$ is the square root $\sqrt{A}$ of such an expression A. But the number of the expressions (13) is 2^k, and by adding all the equations so formed there results

$$2^k \operatorname{sn}\Omega = \sum \sqrt{A}. \tag{15}$$

It remains to be observed that one half of the quantities ψ, and hence also of the A, vanish. For

$$-1 \equiv cc' \cdots \pmod{n},$$

and if one therefore puts $\alpha = -1$ in (14), it follows that

$$\psi(-\Omega) = (\pm 1)(\pm 1) \cdots \psi(\Omega),$$

whereas on the other hand

$$\psi(-\Omega) = -\psi(\Omega).$$

Consequently $\psi(\Omega)$ always vanishes when among the k quantities ± 1 occurring in (14) an even number of negative units is contained.

The roots $x_{\mu,\mu'}$ can therefore be expressed linearly by 2^{k-1} square roots.

If n is a prime number or a power of a prime number, then the squares of the roots of the division equation are rationally expressible by the roots of the transformation equation.[2]

If n is a composite number, then the division of the roots $x_{\mu,\mu'}$ into rows can be carried still further. If p is a prime number contained in n, then one takes two roots $x_{\mu,\mu'}$, $x_{\nu,\nu'}$ into the same row or into different rows according as

$$\mu\nu' - \nu\mu' \equiv 0 \pmod{p}$$

or not. If, according to this, $x_{\nu,\nu'}$ and $x_{\nu_1,\nu_1'}$ belong to one row with $x_{\mu,\mu'}$, then they also belong among themselves to the same row. The number of rows so formed is, as is easily shown, $p+1$, and in this way one obtains, as the first resolvent of the division equation, a transformation equation belonging to the divisor p. We shall later show by another path how the transformation equations for composite divisors can be reduced to those for prime divisors.

[2] Compare, on this theorem, Kronecker, Monatsbericht der Berliner Akademie, 19 July 1875. Kronecker there points out an oversight which occurs in a paper of Jacobi concerning this subject (Crelle, vol. 47 and vol. 50).